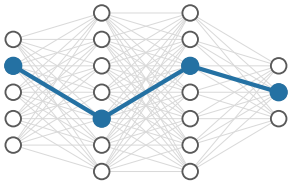
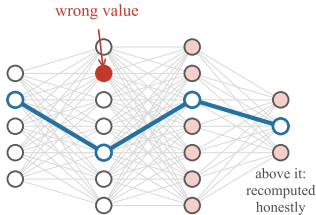


(a) Anchuri et al.: check one random path



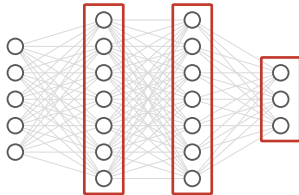
opens ≈ 1 row per layer
misses a node w.p. $1 - 1/N$

(b) Our attack: change one neuron



the path misses it: accepted
with probability $1 - 1/N$ (99.8%)

(c) Our defence: check every layer at once
 $\chi^\top Z \stackrel{?}{=} (\chi^\top W) X$ for every layer



any wrong value is caught
with probability $1 - 2^{-\lambda}$